

Problem Set #8

Problem 1

(i) Eigenstates localized in the left well are

$$\psi_{\ell n}(x) = \begin{cases} \sqrt{\frac{4}{L}} \sin\left(\frac{2\pi n x}{L}\right) & -\frac{L}{2} \leq x < 0 \\ 0 & x > 0 \end{cases}$$

while those localized to the right are.

$$\psi_{r n}(x) = \begin{cases} 0 & x < 0 \\ \sqrt{\frac{4}{L}} \sin\left(\frac{2\pi n x}{L}\right) & 0 \leq x \leq \frac{L}{2} \end{cases}$$

$\psi_{\ell n}, \psi_{r n}$ have the ~~same~~ eigenvalues

$$E_n = \frac{2\pi^2 n^2}{m L^2}$$

Each state is 2-fold degenerate.
The ground state energy is

$$E_0 = \frac{2\pi^2}{m L^2}$$

Denote the degenerate states by $| \ell 0 \rangle$ and $| r 0 \rangle$

(ii) Consider the perturbation

$$H_1 = \begin{cases} V_0 & -a/2 < x < a/2 \\ 0 & \text{otherwise} \end{cases}$$

To calculate the perturbed energy levels need to diagonalize the matrix

$$\begin{bmatrix} \langle l_0 | H_1 | l_0 \rangle & \langle l_0 | H_1 | r_0 \rangle \\ \langle r_0 | H_1 | l_0 \rangle & \langle r_0 | H_1 | r_0 \rangle \end{bmatrix}$$

Consider

$$\langle l_0 | H_1 | r_0 \rangle = V_0 \int_{-a/2}^{a/2} dx \psi_{l_0}^*(x) \psi_{r_0}(x)$$

However this integral vanishes, since in

$$-a/2 < x < 0 \quad \psi_{r_0} = 0$$

$$\text{while } 0 < x < a/2 \quad \psi_{l_0} = 0$$

$$\text{Thus } \psi_{l_0}^*(x) \psi_{r_0}(x) = 0 \quad \text{in} \quad -a/2 \leq x \leq a/2$$

$$\text{Similarly } \langle r_0 | H_1 | l_0 \rangle = 0$$

Thus H_1 is already diagonal in the degenerate subspace.

Eigenvalues of H_1 are

$$\begin{aligned} E_l^{(1)} &= V_0 \int_{-a/2}^0 dx \psi_{l0}^*(x) \psi_{l0}(x) \\ &= \frac{V_0 a}{4} - \frac{V_0 L}{8\pi} \sin\left(\frac{2\pi a}{L}\right) \equiv \bar{E}^{(1)} \end{aligned}$$

$$\begin{aligned} E_r^{(1)} &= V_0 \int_0^{a/2} dx \psi_{r0}^*(x) \psi_{r0}(x) \\ &= \frac{V_0 a}{2} - \frac{V_0 L}{8\pi} \sin\left(\frac{2\pi a}{L}\right) = \bar{E}^{(1)} \end{aligned}$$

Thus degeneracy is not lifted

~~(iv)~~ For $H_2 = \begin{cases} -V_0 & -\frac{a}{2} \leq x \leq 0 \\ V_0 & 0 \leq x \leq a/2 \end{cases}$

We clearly have

$$E_l^{(1)} = -\bar{E}^{(1)}$$

$$E_r^{(1)} = \bar{E}^{(1)}$$

Thus new energy levels are

$$\frac{2\pi^2}{mL^2} \pm \bar{E}^{(1)} \quad \text{degeneracy is lifted}$$

iii
(iv) For either perturbation there cannot be any degeneracy in the exact solution

Suppose in a 1dim QM problem there are two states $\psi_1(x)$, $\psi_2(x)$ with the same energy.

Then

$$\frac{d^2\psi_1}{dx^2} + 2m(E - V(x))\psi_1 = 0$$

$$\frac{d^2\psi_2}{dx^2} + 2m(E - V(x))\psi_2 = 0$$

Then the Wronskian W is a constant

$$W = \psi_1 \frac{d\psi_2}{dx} - \psi_2 \frac{d\psi_1}{dx}$$

Since this is a constant we may evaluate this at any value of x , eg at $x = L/2$. Here $W=0$

Thus

$$\frac{\psi_1'}{\psi_1} = \frac{\psi_2'}{\psi_2}$$

$\Rightarrow$

$$\psi_1(x) \propto \psi_2(x)$$

$\Rightarrow$ They are not

Problem 2:

There are four states with $n=2$ which are degenerate. Denote these by

$$|1\rangle = |2, 1, 1\rangle$$

$$|2\rangle = |2, 1, -1\rangle$$

$$|3\rangle = |2, 1, 0\rangle$$

$$|4\rangle = |2, 0, 0\rangle$$

where the notation is $|n, l, m\rangle$.

The perturbation to the H-atom hamiltonian is

$$H_1 = eEz$$

Clearly

$$\langle nlm | z | nlm \rangle = 0$$

by parity considerations

Now consider

$$\langle nlm | z | n l' m' \rangle$$

Since $[z, L_z] = 0$, the state

$$L_z (z | n l' m' \rangle) = m' (z | n l' m' \rangle)$$

Since states with different L_z have no overlap

$$\langle n l m | z | n l' m' \rangle = A_{l, l'} B_m \delta_{mm'}$$

This means that the nonzero matrix elements for $n=2$ are

$$\langle 3 | H_1 | 4 \rangle = \langle 4 | H_1 | 3 \rangle^*$$

$$\langle 3 | H_1 | 4 \rangle = eE \int dr r^2 \sin \theta \int d\theta d\phi u_{210}^*(r, \theta, \phi) \cos \theta u_{200}(r, \theta, \phi)$$

Using

$$u_{210}(r, \theta, \phi) = \left(\frac{1}{32\pi a_0^3} \right)^{1/2} \frac{r}{a_0} e^{-r/2a_0} \cos \theta$$

$$u_{200}(r, \theta, \phi) = \left(\frac{1}{32\pi a_0^3} \right)^{1/2} \left(2 - \frac{r}{a_0} \right) e^{-r/2a_0}$$

we get

$$\langle 3 | H_1 | 4 \rangle = \frac{2\pi eE}{32\pi a_0^3} \frac{1}{a_0} \int dr r^4 \left(2 - \frac{r}{a_0} \right) e^{-r/a_0} \int_0^\pi d\theta \sin \theta \cos^2 \theta$$

$$= - \frac{eE}{16a_0^4} 72 a_0^5 \cdot \frac{2}{3} = -3eEa_0$$

$$\langle 4 | H_1 | 3 \rangle = -3eEa_0$$

In usual degenerate perturbation theory we need to start with a state

$$|\psi\rangle = \sum_{i=1}^4 a_i |i\rangle$$

Then the first order perturbation theory leads to the equations

$$\sum_j [\langle j | H_1 | i \rangle - E^{(1)} \delta_{ij}] a_j = 0$$

where $E^{(1)}$ is the correction
This gives

$$0 = \begin{pmatrix} -E^{(1)} & 0 & 0 & 0 \\ 0 & -E^{(1)} & 0 & 0 \\ 0 & 0 & -E^{(1)} & -3eEa_0 \\ 0 & 0 & -3eEa_0 & -E^{(1)} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}$$

Thus

$$(E^{(1)})^2 ((E^{(1)})^2 - (3eEa_0)^2) = 0$$

Thus

$$E^{(1)} = 0, 0, 3eEa_0, -3eEa_0$$

Thus the degeneracy is only partly lifted to this order.

Problem 3:

(i) The unperturbed wavefunctions are plane waves

$$\psi_{\vec{k}}(\vec{x}) = \frac{1}{\sqrt{L^3}} e^{i\vec{k} \cdot \vec{x}}$$

Since $\psi_{\vec{k}}(\vec{x})$ is periodic we must have

$$\vec{k} = \frac{2\pi\vec{n}}{L}$$

where $\vec{n}$ is an integer valued vector

(ii) Since $V(\vec{x} + a\vec{e}_x) = V(\vec{x})$

$$\vec{g} = \frac{\vec{m}}{a}$$

where $\vec{m}$ is an integer valued vector.

Clearly this is consistent with periodicity across the box, since

$$e^{2\pi i \vec{g} \cdot \vec{x}} = e^{\frac{2\pi i \vec{m} \cdot \vec{x}}{a}} = e^{2\pi i N \vec{m} \cdot \vec{x}} = 1$$

(iii) The matrix element of the perturbation is

$$\begin{aligned}\langle \vec{k}' | V | \vec{k} \rangle &= \sum_{\vec{g}} V(\vec{g}) \frac{1}{L^3} \int d^3x e^{i(\vec{k} - \vec{k}' + 2\pi\vec{g}) \cdot \vec{x}} \\ &= \sum_{\vec{g}} V(\vec{g}) \delta(2\pi\vec{g} - \vec{k}' + \vec{k}) \\ &= V\left(\frac{1}{2\pi}(\vec{k}' - \vec{k})\right) \equiv V(\vec{g}_0).\end{aligned}$$

From usual perturbation theory the first order correction to the state $|\vec{k}\rangle$ is

$$\frac{1}{\sqrt{L^3}} \sum_{\vec{k}' \neq \vec{k}} \frac{\langle \vec{k}' | V | \vec{k} \rangle}{E_k - E_{k'}} |\vec{k}'\rangle$$

$$\begin{aligned}\psi_k^{(1)}(\vec{x}) &= \frac{1}{\sqrt{L^3}} \sum_{\vec{g} \neq 0} \frac{V(\vec{g})}{k^2 - (\vec{k} + 2\pi\vec{g})^2} \cdot \frac{2m}{\hbar^2} e^{i(\vec{k} + 2\pi\vec{g}) \cdot \vec{x}} \\ &= \frac{1}{\sqrt{L^3}} \frac{2m}{\hbar^2} \sum_{\vec{g} \neq 0} \frac{V(\vec{g}) e^{i(\vec{k} + 2\pi\vec{g}) \cdot \vec{x}}}{k^2 - (\vec{k} + 2\pi\vec{g})^2}\end{aligned}$$

The first order correction to the energy is

$$E_k^{(1)} = \langle \vec{k} | V | \vec{k} \rangle = V(0).$$

The second order correction is

$$\frac{2m}{\hbar^2} \sum_{\vec{g} \neq 0} \frac{|V(\vec{g})|^2}{\vec{k}^2 - (\vec{k} + 2\pi\vec{g})^2}$$

This works for generic $\vec{k}$.

(iv) when $\vec{k}^2 = (\vec{k} + 2\pi\vec{g})^2$
 i.e. $0 = 4\pi \vec{g} \cdot \vec{k} + 4\pi^2 \vec{g}^2$

This breaks down since the unperturbed states with $\vec{k}, \vec{k} + 2\pi\vec{g}$ are degenerate

We then need to choose the correct linear combination which diagonalizes. Let this be

$$|\psi\rangle = a|\vec{k}\rangle + b|\vec{k} + 2\pi\vec{g}\rangle$$

Then

$$0 = \begin{bmatrix} \langle \vec{k} | V | \vec{k} \rangle - E^{(1)} & \langle \vec{k} | V | \vec{k} + 2\pi\vec{g} \rangle \\ \langle \vec{k} + 2\pi\vec{g} | V | \vec{k} \rangle & \langle \vec{k} + 2\pi\vec{g} | V | \vec{k} + 2\pi\vec{g} \rangle - E^{(1)} \end{bmatrix}$$

ie

$$\det \begin{bmatrix} V_0 - E^{(1)} & V(\vec{g}) \\ V^*(\vec{g}) & V_0 - E^{(1)} \end{bmatrix} = 0$$

$$\Rightarrow (V_0 - E^{(1)})^2 = |V(\vec{g})|^2$$

$$E^{(1)} = V_0 \pm |V(\vec{g})|$$

The wave fns are

$$\psi_{\pm} = \frac{1}{\sqrt{2L^3}} \left[e^{i\vec{k} \cdot \vec{x}} \pm e^{i(\vec{k} + 2\pi\vec{g}) \cdot \vec{x}} \right]$$

Problem 34 (18.2.1)

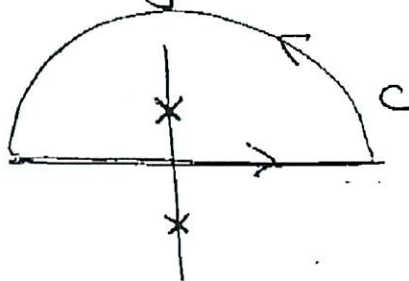
$$H'(t) = -\frac{eEx}{1+(t/\tau)^2}$$

Then

$$d_u(\infty) = -\frac{1}{\hbar} (-eE) \int_{-\infty}^{\infty} dt e^{i\omega t n} \frac{\langle u|x|0 \rangle}{1+(t/\tau)^2}$$

$$\begin{aligned} d_1(\infty) &= \frac{ieE}{\hbar} \left(\frac{\hbar}{2m\omega} \right)^{1/2} \int_{-\infty}^{\infty} \frac{dt e^{i\omega t}}{1+(t/\tau)^2} \\ &= \frac{ieE}{\hbar} \left(\frac{\hbar}{2m\omega} \right)^{1/2} \tau \int_{-\infty}^{\infty} \frac{d\eta e^{i\nu\eta}}{1+\eta^2} \quad (\nu = \omega\tau) \end{aligned}$$

For $\nu > 0$ the integral may be evaluated by closing contour in UHP.

$$\int_C \frac{d\eta e^{i\nu\eta}}{1+\eta^2} =$$


$$= \frac{2\pi i}{2i} e^{-\nu} = \pi e^{-\omega\tau}$$

Thus

$$d_1(\infty) = \frac{ieE}{\hbar} \left(\frac{\hbar}{2m\omega} \right)^{1/2} \pi \tau e^{-\omega\tau}$$

$\Rightarrow$

$$P_{0 \rightarrow 1} = \frac{e^2 E^2 \pi^2 \tau^2}{2m\omega\hbar} e^{-2\omega\tau}$$

Problem 5 (18.2.3)

For a particle in a box of size L ,
the ground state energy is

$$E_0 = \frac{\pi^2 \hbar^2}{2mL^2}$$

Thus the ~~max~~ average momentum is

$$p = \sqrt{2mE} = \frac{\pi \hbar}{L}$$

Time taken by particle to go across
box is

$$T = \frac{Lm}{p} = \frac{mL^2}{\pi \hbar}$$

This is the time scale of problem.

If box is increased to $2L$

in a time $\tau \ll T$, the system
remains in ground state of the
original box, i.e., at time $t=0$
when the change is made

$$\psi(x,0) = \begin{cases} \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ 0 & \frac{L}{2} \leq x \leq L \\ & -L \leq x \leq -\frac{L}{2} \end{cases}$$

The ground state wavefn in box of size $2L$ is

$$\psi'(x) = \sqrt{\frac{1}{L}} \cos\left(\frac{\pi x}{2L}\right) \quad -L \leq x \leq L$$

Thus the overlap is

$$\frac{\sqrt{2}}{L} \int_{-L/2}^{L/2} dx \cos\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{2L}\right) = \frac{8}{3\pi}$$

$$\text{Thus } P_{\text{ub}} = \left(\frac{8}{3\pi}\right)^2$$